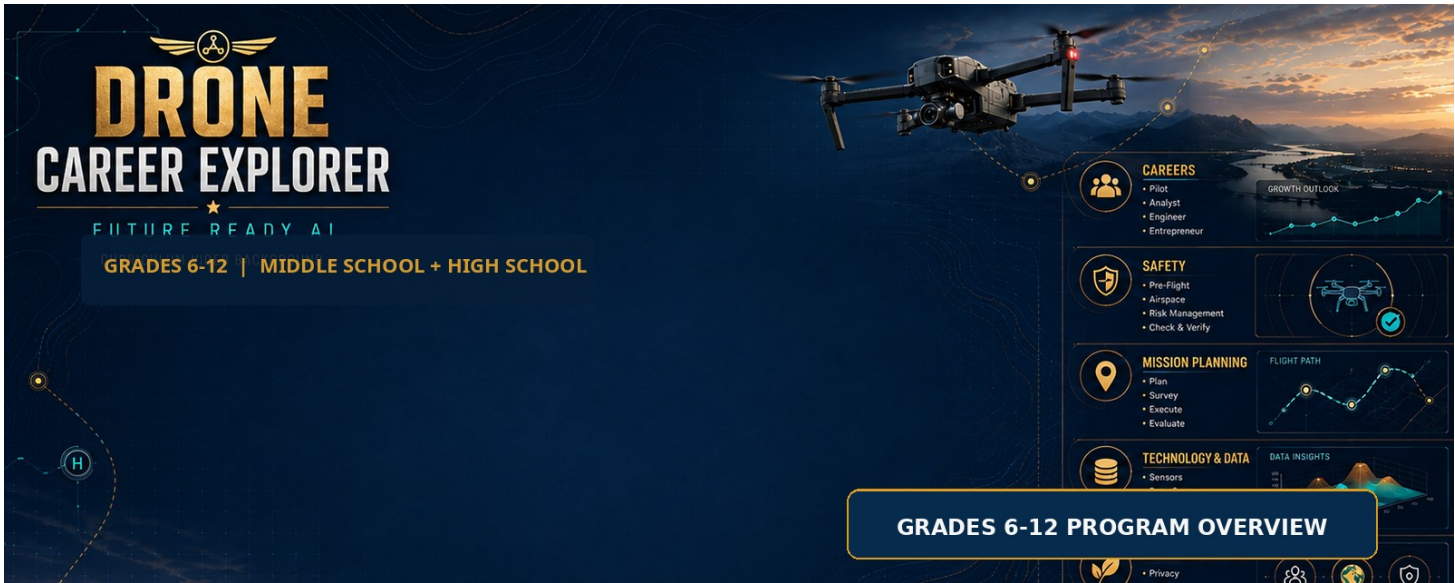




DRONE CAREER EXPLORER: GRADES 6-12

PROGRAM OVERVIEW

A career-connected pathway that helps students understand drone-enabled systems, responsible innovation, emerging industries, and future opportunities - while strengthening the human judgment needed to use technology well.

GRADES	PATHWAYS	MIDDLE SCHOOL	HIGH SCHOOL	CULMINATION
6-12	2	6 units	9 units	Capstone + Portfolio

PROGRAM PURPOSE & VISION

Drone Career Explorer introduces students to drones as part of larger systems - not as isolated flying devices. Learners investigate how people, technology, data, workflows, safeguards, and mission purpose combine to create useful and responsible solutions. The pathway is designed to move students from awareness to applied reasoning, from career curiosity to portfolio evidence, and from technology fascination to thoughtful decision-making.

MIDDLE SCHOOL GRADES 6-8	HIGH SCHOOL GRADES 9-12
EXPLORE THE SYSTEM • Build foundational understanding of drone systems, safety, data, and mission purpose. • Strengthen systems thinking, teamwork, responsible-use habits, and career awareness. • Culminate in the Drone Career Challenge - a responsible solution concept for a real-world need.	APPLY THE SYSTEM • Analyze mission planning, AI and sensors, aerial data, logistics, industry workflows, and human accountability. • Strengthen evidence-based reasoning, workforce readiness, communication, and career portfolio development. • Culminate in a Capstone Mission Proposal that students research, design, communicate, and defend.

WHAT MAKES THE PATHWAY DIFFERENT

CAREER-CONNECTED	NO-FLIGHT-READY	EVIDENCE-DRIVEN	HUMAN-CENTERED
Every unit links learning to industries, roles, and future pathways.	Rigorous learning can occur through cases, planning, simulation, and evidence analysis.	Students explain decisions using information, comparison, and reasoning.	Safety, privacy, responsibility, and human judgment remain central.

THE PROGRAM PROMISE

Students move beyond asking “How do drones work?” and begin asking: What problem are we solving? What evidence do we need? What safeguards are necessary? Who remains responsible? Where could I fit into this future?

HOW THE GRADES 6-12 PATHWAY PROGRESSES

The pathway intentionally increases complexity across grade bands. Middle school establishes the language, habits, and systems awareness students need; high school asks students to apply those ideas to more complex evidence, industry, workforce, and capstone decisions.

LEARNING DIMENSION	GRADES 6-8	GRADES 9-12
Systems Thinking	Recognize how people, technology, data, process, and purpose connect.	Analyze complex workflows, constraints, stakeholders, and accountability.
Safety & Responsibility	Identify privacy, safety, and responsible-use considerations.	Evaluate ethics, public trust, human oversight, and operational safeguards.
Data & Evidence	Understand that drones collect information that people interpret.	Judge evidence quality, compare data, and defend claims and recommendations.
Mission Planning	Plan a simplified mission around purpose, people, information, and safeguards.	Develop mission concepts with constraints, success criteria, evidence, and workflow.
Industry & Careers	Explore how drones support multiple industries and roles.	Connect applications to workforce pathways, professional skills, and business needs.
Student Evidence	Complete activities, reflections, and a Drone Career Challenge.	Build portfolio evidence and defend a Capstone Mission Proposal.

CURRICULUM SEQUENCE

MIDDLE SCHOOL | 6 UNITS

- 1. Meet the Drone Ecosystem
- 2. Safety Before Flight
- 3. How Drones See and Sense
- 4. Mission Planning Basics
- 5. Drones at Work
- 6. Drone Career Challenge

HIGH SCHOOL | 9 UNITS

- 1. Drone Systems & Emerging Careers
- 2. Safety, Responsibility & Human Accountability
- 3. Mission Planning & Operations
- 4. AI, Sensors & Autonomous Systems
- 5. Aerial Data & Evidence
- 6. Industry Mission Lab
- 7. Logistics & Autonomous Delivery
- 8. Workforce Readiness
- 9. Capstone Mission Proposal

CORE LEARNING THEMES

SYSTEMS + TECHNOLOGY How tools, people, workflows, and purpose connect.	AI, SENSORS + DATA How information is collected, interpreted, evaluated, and used.	MISSION PLANNING How goals, constraints, stakeholders, and evidence shape solutions.
SAFETY + RESPONSIBILITY How ethics, privacy, safety, accountability, and human judgment guide use.	INDUSTRIES + CAREERS How drone-enabled work supports real sectors and emerging roles.	PORTFOLIO + CAPSTONE How students communicate learning, reflection, evidence, and future pathways.

FUTURE READY AI LEARNING MODEL

LAUNCH	EXPLORE	EXPLAIN	CREATE	REFLECT	EXTEND
--------	---------	---------	--------	---------	--------

Future Ready AI competencies reinforced across the pathway: UNDERSTAND | EVALUATE | ANALYZE | PROTECT | APPLY

INDUSTRIES & CAREER CONNECTIONS

Drone Career Explorer treats career awareness as part of the learning - not an add-on. Students encounter industries and roles while analyzing systems, evidence, responsibility, mission needs, and professional communication.

CONSTRUCTION progress documentation, inspection	REAL ESTATE imagery, property documentation	LOGISTICS routing, movement, delivery systems	INFRASTRUCTURE inspection, monitoring, maintenance	MEDIA storytelling, coverage, production
MAPPING site information, spatial evidence	ENVIRONMENT monitoring, observation, analysis	PUBLIC SERVICE community applications, response	TECH + DATA analytics, workflows, automation	BUSINESS entrepreneurship, operations, client needs

STUDENT OUTCOMES

SYSTEMS THINKING Explain how people, tools, data, process, and purpose interact.	RESPONSIBLE JUDGMENT Recognize safety, privacy, ethics, and human accountability.	EVIDENCE USE Interpret information and support decisions with reasoning.	MISSION DESIGN Frame a problem and build a responsible solution concept.
CAREER AWARENESS Identify industries, roles, and future pathways.	COLLABORATION Communicate, share responsibility, and work productively.	PORTFOLIO EVIDENCE Curate artifacts that demonstrate growth and capability.	REFLECTION Connect learning to interests, strengths, and future possibilities.

RESOURCE SYSTEM

1 INTRO VIDEO Orient	2 FACILITATOR GUIDE Plan	3 STUDENT WORKBOOK Engage	4 ASSESSMENT + CAPSTONE Assess	5 IMPLEMENTATION GUIDE Launch
--	--	---	--	---

IMPLEMENTATION OPTIONS

FULL PATHWAY Semester / Extended Sequence Teach the grade-band sequence from orientation through capstone and portfolio.	SELECTED MODULES Targeted Learning Use individual units for STEM, CTE, technology, career exploration, or interdisciplinary integration.	PILOT / WORKSHOP Short Implementation Use curated lessons, cases, and career challenges for a pilot, summer program, or enrichment experience.
NO-FLIGHT-READY IMPLEMENTATION Students do not need to physically operate a drone to complete the curriculum. Rigorous learning can occur through case studies, videos, diagrams, simulation, mission planning, image/data analysis, career research, project challenges, and teacher demonstrations. The curriculum is educational and career exploratory; it does not represent FAA certification or professional pilot authorization.		DESIGNED FOR STEM / STEAM CTE TECHNOLOGY ENRICHMENT SUMMER WORKFORCE READINESS

THINK CRITICALLY. DESIGN RESPONSIBLY. LEAD CONFIDENTLY.

Drone Career Explorer connects technology, careers, and real-world problem solving so students can imagine where they fit in the future.